CSTS Labeling & Computing Similarities

PBL:Hugging Face NLP Project
Team3
track 3 & session 3
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Progress Update

- **Automated Data Extraction**: We built a tool using **spaCy** (an NLP library) that automatically reads sentences and picks out the most important parts: the **Action** (the Verb) and the **Goal** (the Object).
    - *Example:* In "He ate an apple," the system identifies **"ate"** as the Action and **"apple"** as the Goal.
- **Smart Scoring System**: We integrated a powerful AI model (**all-MiniLM-L6-v2**) to turn these actions and goals into mathematical vectors. This allows the computer to "calculate" how similar two ideas really are.
- **Data Labeling**: We processed the standard **STSb dataset**, adding our own labels for Actions and Goals to make the evaluation more detailed.

Some Visual Results

A3 | A group of men play soccer on the beach.

	A	B	C	D	E	F	G	H	I	J
1	sentence1	sentence2	sts_score	action1	action2	action_sim	goal1	goal2	goal_sim	
2	A girl is styl	A girl is bru	0.1	style	brush	0.3413	her hair	her hair	1	
3	A group of	men play so	0.144	None	None	1	men	boys	0.6259	
4	One woma	A woman n	0.2	measure	measure	1	another wo	another wo	1	
5	A man is cu	A man is sli	0.167999999	cut	slice	0.713	a cucumber	a cucumber	1	
6	A man is pl	A man is pl	0.06	play	play	1	a harp	a keyboard	0.3024	
7	A woman is	A woman is	0.072	cut	cut	1	onions	tofu	0.2252	
8	A man is ric	A man is ric	0.139999999	rid	rid	1	an electric l	a bicycle	0.8267	
9	A man is pl	A man is pl	0.088	play	play	1	the drums	the guitar	0.5577	
10	A man is pl	A lady is pl	0.088	play	play	1	guitar	the guitar	0.9256	

B3 | A group of boys are playing soccer on the beach.

	A	B	C	D	E	F	G	H	I	J
1	sentence1	sentence2	sts_score	action1	action2	action_sim	goal1	goal2	goal_sim	
2	A girl is styl	A girl is bru	0.1	style	brush	0.3413	her hair	her hair	1	
3	A group of	A group of	0.144	None	None	1	men	boys	0.6259	
4	One woma	A woman n	0.2	measure	measure	1	another wo	another wo	1	
5	A man is cu	A man is sli	0.167999999	cut	slice	0.713	a cucumber	a cucumber	1	
6	A man is pl	A man is pl	0.06	play	play	1	a harp	a keyboard	0.3024	
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- This is part of the data we have labeled
- Current Problem: As illustrated above, some sentences obviously have verbs while the actions output by our model is None

Problems and Causes

1. **Current Code Constraint**

The system currently relies on a strict syntactic filter to identify the primary action:

```
if token.dep_ == "ROOT" and token.pos_ == "VERB":
```

- **Strict AND-Logic:** If the parser misidentifies a verb as a noun (e.g., "dances"), the entire action is lost.
- **Root Dependency:** Fails to capture actions in compound tenses (e.g., "is dancing"), where the auxiliary verb often takes the `ROOT` position.
- **Semantic Loss:** Ignores verbal nouns and compounds that carry the primary intent of the sentence.

Input Sentence	Expected Action	Potential Cause of "None"	
A woman dances in the rain.	**dance**	"Dances" is **tagged as a NOUN** (plural) rather than a VERB, failing the filter.	
A group of men play soccer.	**play**	**Due to complex subjects,** "play" may be misidentified as a NOUN (e.g., a play), excluding it from the rule.	
The ballerina is dancing.	**dance**	The model is biased by "ballerina," interpreting "dancing" as a NOUN/ADJ associated with the profession	
A woman is water skiing.	**ski**	"Water skiing" is treated as a single **Compound Noun,** missing the underlying verbal action.	
Someone in a gorilla costume is playing drums.	**play**	**Long subjects** cause dependency shifts; the break command stops the search before finding the Action.	

Questions

- Should we redefine the Goal more broadly as **'The state of the environment to be achieved by the action,'** thereby including locations and complements (attr) in our extraction and evaluation?

- Are there specific advantages or applications to strictly using the 'Direct Object' as the Goal in this research context?

Current framework:
・extracting the **Direct Object (dobj)** as the 'Goal'

・considering application for real-world systems like robot instructions, the true 'Goal' is often the **Final State (State)** of the environment.